

VNUHCM - University of Science
Faculty of Information Technology

CS486

Introduction to Database Systems

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Timetable

■ Lecture

- May 14 ~ Aug 08, 2024 (Jun 24 ~ Jul 13: off)
- Room I41
 - 22TT1: Tue & Thu, 15:30~17:10
 - 22TT2: Tue & Thu, 13:30~15:10
- Office hour: Sat, 14:00~17:00
 - Please make an appointment via email

■ Lab

- TAs
 - Vũ Thị Mỹ Hằng, PhD. (vtmhang@fit.hcmus.edu.vn)
 - Hồ Thị Hoàng Vy, MSc. (hthvy@fit.hcmus.edu.vn)
 - Phan Thị Phương Uyên, MSc. (ptpuyen@fit.hcmus.edu.vn)
- Time

Objective

- Describe
 - Roles of the database in an organization
 - Basic concepts of the database system
- Apply common data models in data modeling
- Master query languages to exploit the relational database
- Explain concepts of transactions, scheduler and locking, and recovery management
- Learn skills of
 - Defining schemas & building queries in a DBMS
 - Critical thinking in practice

Learning outcome

- (1) Interpret *roles of the database* in organizations
- (2) Describe *basic concepts* of the database
- (3) Draw *the entity relationship model (ER)* at a basic level
- (4) Interpret concepts of *the relational data model*
- (5) Master of *query languages*
- (6) Use a DBMS proficiently (*MS SQL Server* or *PostgreSQL*) to
 - Declare schemas of the relational database
 - Exploit the data

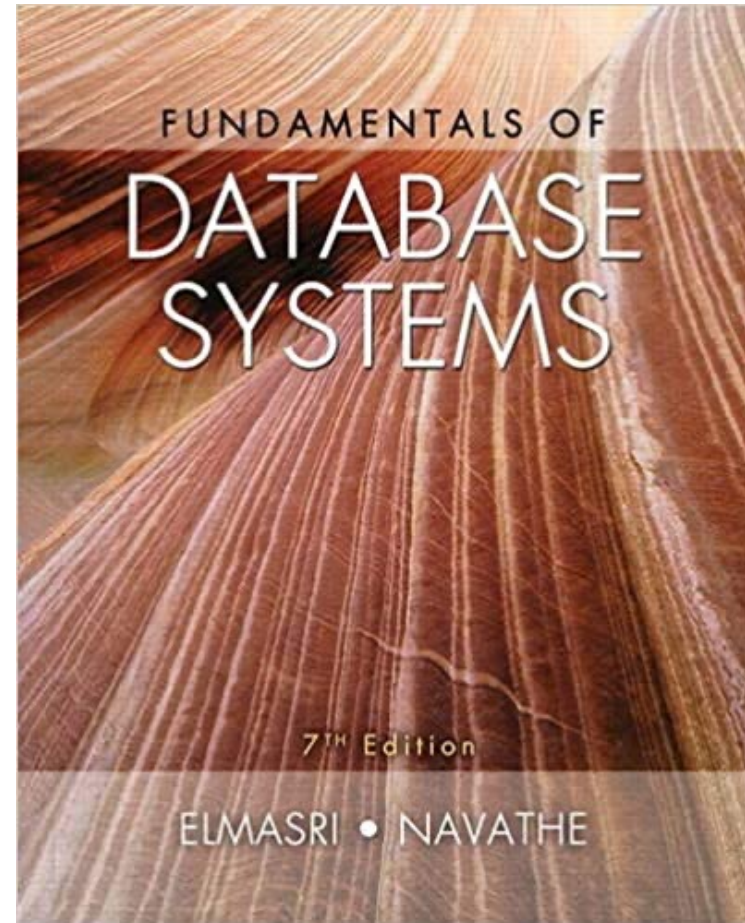
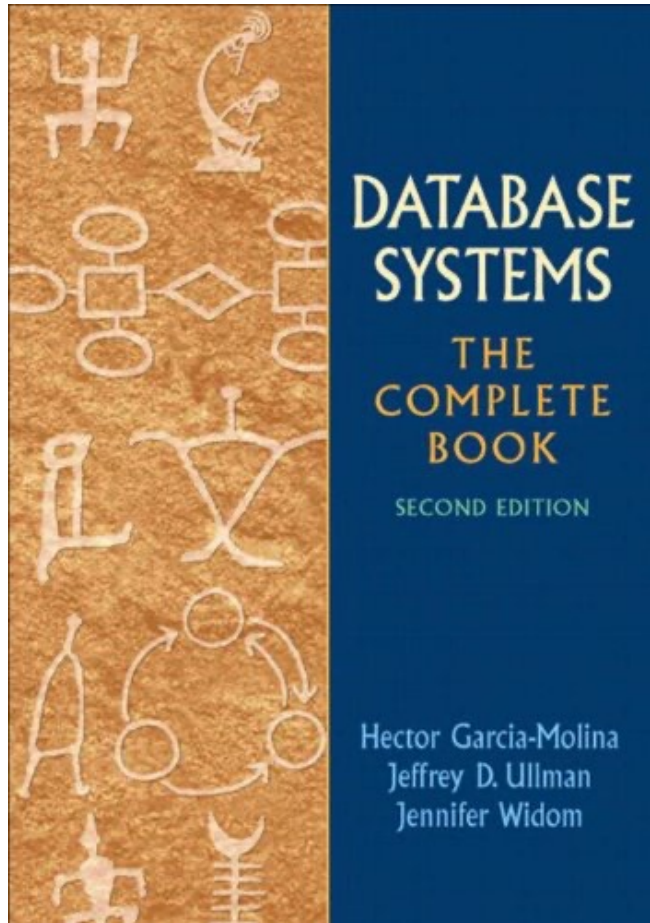
Learning outcome

- (7) Detect ***integrity constraints*** in the relational database
- (8) Categorize concepts of ***functional dependency and normal forms***
- (9) Explain how to ***transaction, scheduler, locking, and data recovery*** work
- (10) Describe ***future trends*** of the database

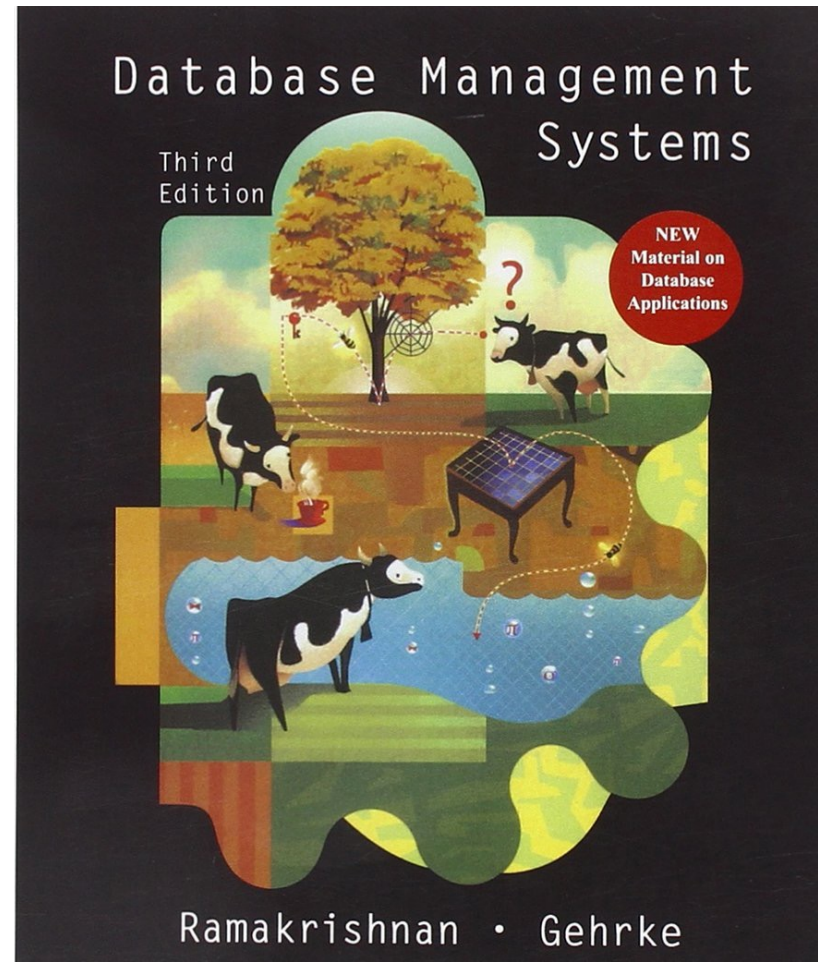
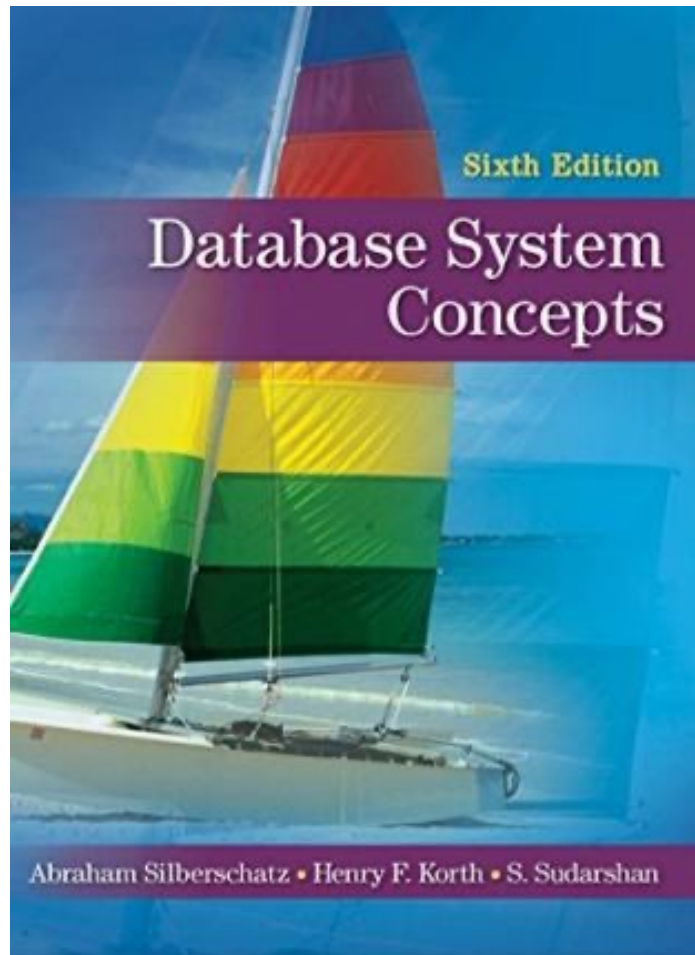
Content

- 01- Introduction
- 02- Entity relationship model
- 03- Relational data model
- 04- Query languages
- 05- Integrity constraint
- 06- Functional dependency and normal forms
- 07- Transaction and scheduler
- 08- Recovery management
- 09- Other issues

Textbook



Reference



Resource

- Moodle
 - Slide, assignment, lab document
 - Announcement, Q&A
- Google drive

Grade

- **1. Lecture (40%)**
 - Final exam: writing
 - Closed book, 90 minutes

- **2. Assignment (20%)**
 - In-class
 - Homework

- **3. Lab (40%)**
 - Many quizzes, midterm & final exams
 - Closed book

Study method

- Flipped Classroom : 3+ weeks
- At home (self study)
 - Watching video
 - Reading book
- In class
 - Take quizzes
 - Discussion
 - Q&A

Rule

- Assignment
 - Has to be done over 70%, grade will be counted
 - Is submitted late or does not follow the format → 0 point
- Cheating → fail
- If one of the three component scores is 0
 - The total grade will be 0

Rule (cont.)

■ In class

- Keep quiet (do not chat)
- Switch your cell phone to silent mode
- Raise your hand if you have comments/questions
- Be on time
- Do not bring food & beverage
- Well dressed

■ If you break the rules

- Must quit the class
- Be marked “absent” from class

Cheating: Description

- **Please pay close attention, especially if this is your first semester at CMU**
- What is cheating?
 - Sharing code: by copying, retyping, **looking at**, or supplying a file
 - Describing: verbal description of code from one person to another.
 - Coaching: helping your friend to write a lab, line by line
 - Searching the Web for solutions
 - Copying code from a previous course or online solution
 - You are only allowed to use code we supply, or from the CS:APP website
- What is NOT cheating?
 - Explaining how to use systems or tools
 - Helping others with high-level design issues
- See the course syllabus for details.
 - Ignorance is not an excuse

